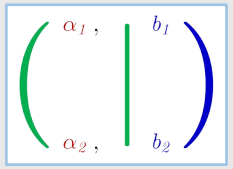
Model Kernelization



Normal Model

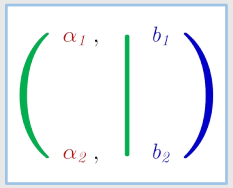
Calculate $a^{\mathrm{T}} b$

$$\begin{bmatrix} \alpha_1 \alpha_2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

O/P

Map Input data to higher dimensions

Before Apply to the model



Vectors Mapping

$a, b \Rightarrow u, v$

Higher Computational Complexity !!!

$$\begin{bmatrix} u_1 , v_1 \\ u_2 , v_2 \\ u_3 , v_3 \\ \vdots , \vdots \\ u_n , v_n \end{bmatrix}$$

Kernelized Model

Calculate $u^{\mathrm{T}} v$

$u_1\ u_2\ u_3 \cdots \cdots u_n$

v_1
v_2
v_3
$\vdots$
v_n

Higher Computational Complexity !!!

O/P